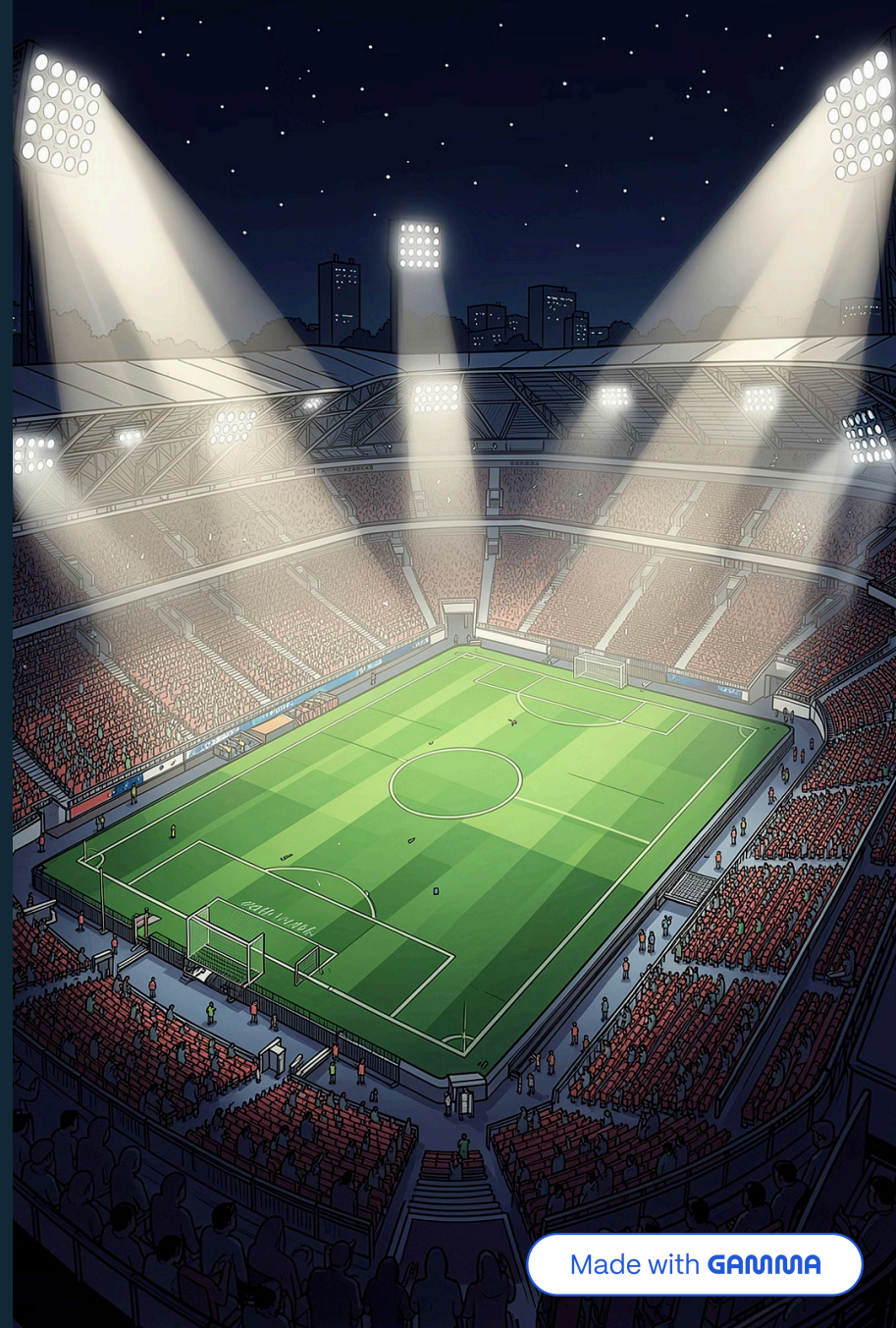
FIFA 21 Player Market Analysis

Data Cleaning · Exploratory Data Analysis · Visualization · Business Intelligence

PYTHON · PANDAS · MATPLOTLIB · POWER BI



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The Business Problem

Football clubs spend **millions of euros** every transfer window. Identifying talented players with high future potential while maintaining financial efficiency remains a complex, high-stakes challenge.

The Challenge

Transfer decisions are often driven by intuition rather than evidence, leading to costly mistakes and inefficient spending.

The Approach

This project applies Python-based data cleaning, EDA, and visualization to FIFA 21 player data — enabling smarter, evidence-based recruitment.

The Outcome

Delivers actionable business insights, investment metrics, and an interactive Power BI dashboard for transfer decision support.



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Dataset Overview



FIFA 21 Raw Dataset

A comprehensive player database covering professional footballers worldwide — the foundation for all analytical work in this project.

- **18,979 players** across global leagues and competitions
- **70+ features** spanning performance, financial, and physical attributes
- Market values, wages, ratings, positions, and nationalities
- Raw dataset :

	ID	Name	LongName	photoUrl	playerUrl	Nationality	Age	↓OVA	POT
0	158023	L. Messi	Lionel Messi	https://cdn.sofifa.com/players/158/023/21_60.png	http://sofifa.com/player/158023/lionel-messi/2...	Argentina	33	93	93
1	20801	Cristiano Ronaldo	C. Ronaldo dos Santos Aveiro	https://cdn.sofifa.com/players/020/801/21_60.png	http://sofifa.com/player/20801/c-ronaldo-dos-s...	Portugal	35	92	92
2	200389	J. Oblak	Jan Oblak	https://cdn.sofifa.com/players/200/389/21_60.png	http://sofifa.com/player/200389/jan-oblak/210006/	Slovenia	27	91	93
3	192985	K. De Bruyne	Kevin De Bruyne	https://cdn.sofifa.com/players/192/985/21_60.png	http://sofifa.com/player/192985/kevin-de-bruyn...	Belgium	29	91	91
4	190871	Neymar Jr	Neymar da Silva Santos Jr.	https://cdn.sofifa.com/players/190/871/21_60.png	http://sofifa.com/player/190871/neymar-da-silv...	Brazil	28	91	91

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Data Cleaning Pipeline

A rigorous, multi-stage workflow transformed the raw FIFA 21 dataset into a reliable, analysis-ready foundation.



Engineered Business Features

Potential Growth	Wage-to-Value Ratio	Skill per Euro	Contract Year

THE CLEANED DATASET :

```
Sample of Newly Cleaned Columns:
                Name  Skill_Moves  Height  Weight Contract_Valid_Until
0           L. Messi            4     170      72                 2021
1  Cristiano Ronaldo            5     187      83                 2022
2           J. Oblak            1     188      87                 2023
3       K. De Bruyne            4     181      70                 2023
4          Neymar Jr            5     175      68                 2022
```

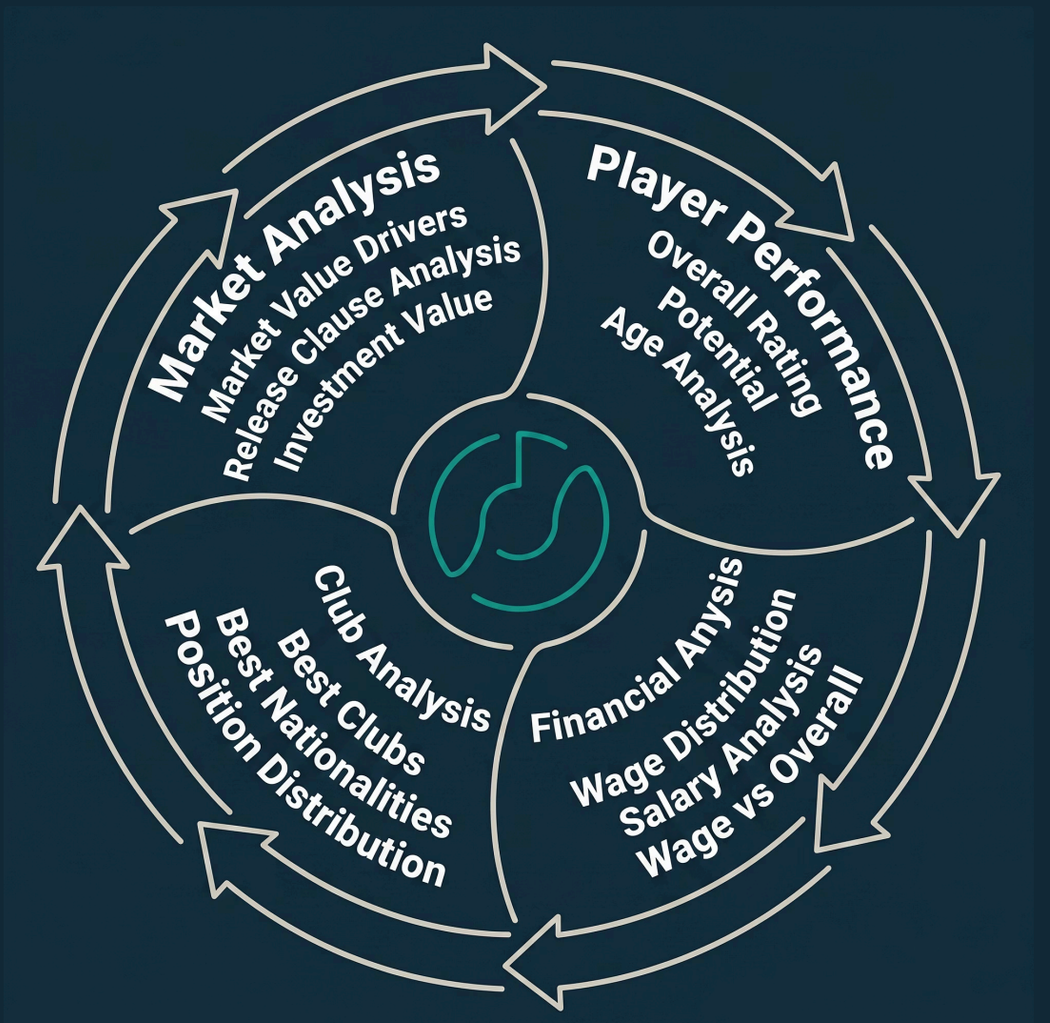
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Exploratory Data Analysis

Four Analytical Pillars

Each dimension of the analysis addresses a distinct business question — from player valuation to squad composition and financial efficiency.

- **Market Analysis** — Value drivers and investment potential
- **Player Performance** — Ratings, potential, and age dynamics
- **Financial Analysis** — Wage distribution and salary efficiency
- **Club Analysis** — Top clubs, nationalities, and positions



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Key Insights



Rating Drives Value

Overall rating is the strongest predictor of market value — a direct, quantifiable relationship.



Wages Reflect Performance

High-rated players command significantly higher wages, validating performance-based compensation.



Youth Is the Best Investment

Young players with high potential deliver the strongest long-term return on investment.



Hidden Gems Exist

Undervalued players with high potential represent low-risk, high-reward transfer opportunities.



Wage-to-Value Efficiency

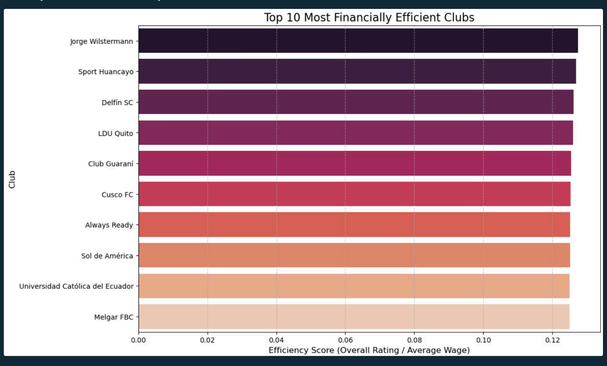
The Wage-to-Value Ratio identifies financially efficient signings, reducing transfer risk.

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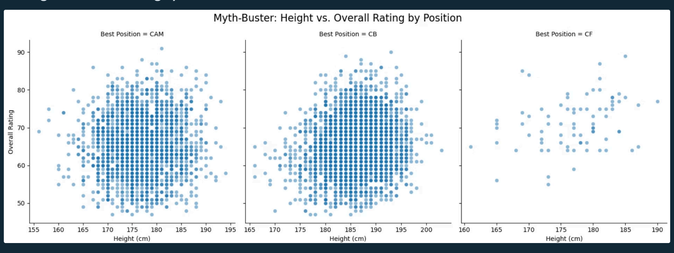
Data Visualizations

Ten core visualizations form the analytical backbone of this project, revealing patterns in player value, performance, and financial efficiency.

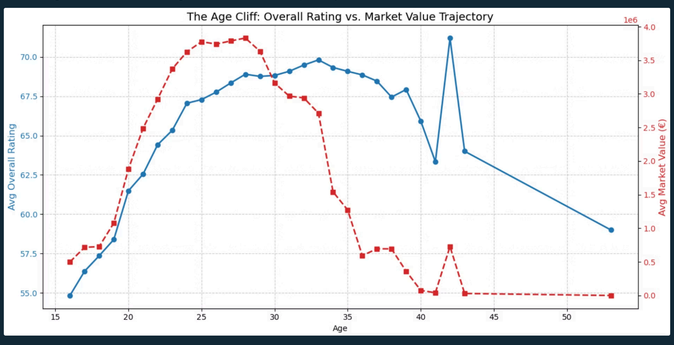
1. Top 10 Most Financially Efficient Clubs



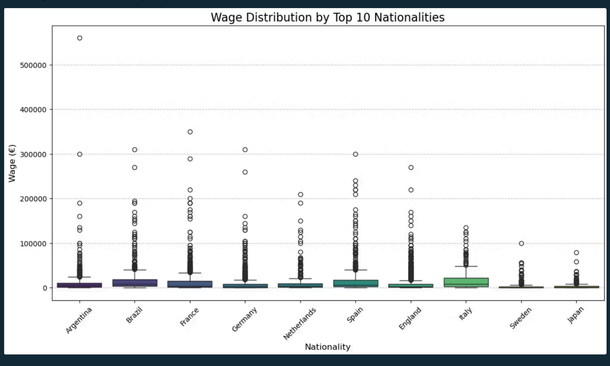
2. Height vs Overall Rating by Position



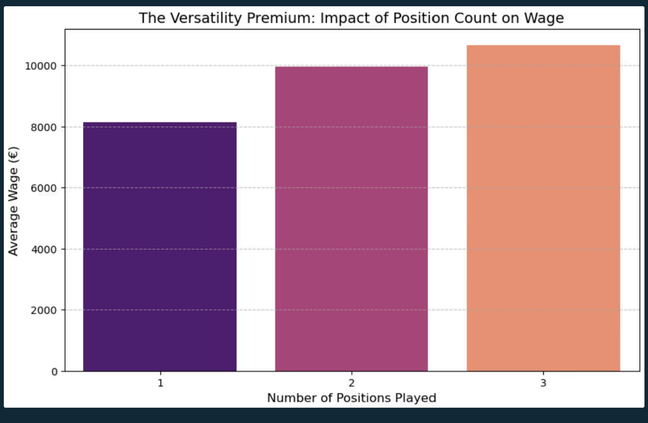
3. The Age Cliff: Overall Rating vs Market Value Trajectory



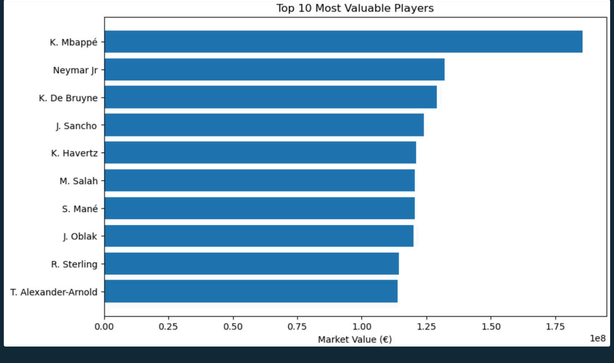
4. Wage Distribution By Top 10 Nationalities



5. The Versatility Premium : Impact of Position Count on Wage

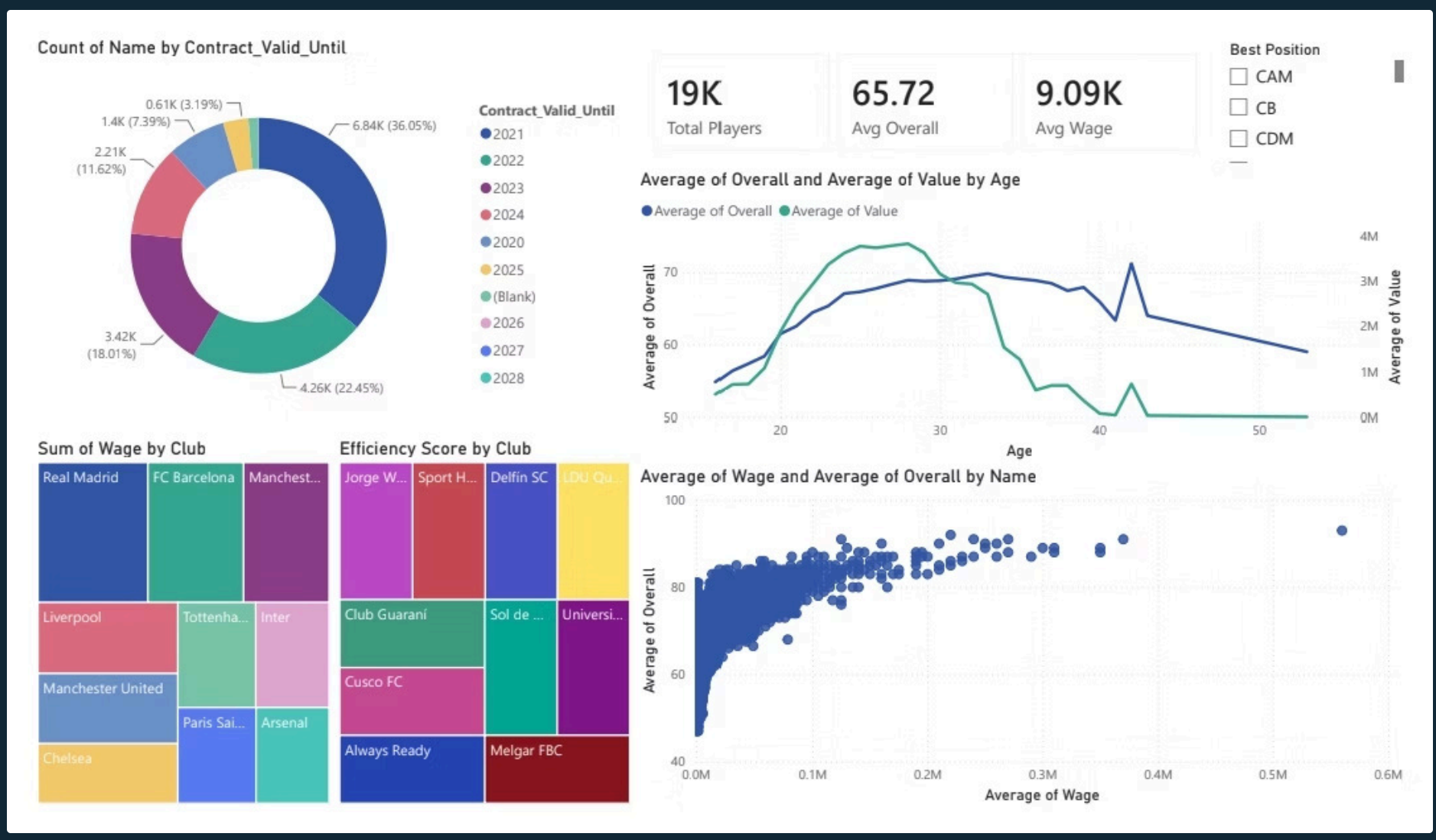


6. Top 10 Most Valuable Players



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Power BI Interactive Dashboard





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Business Recommendations

Invest in Young, High-Potential Players

Prioritize players under 23 with high potential ratings for maximum long-term value and resale potential.

Monitor Wage-to-Value Ratio

Use this metric before every signing to ensure financial efficiency and avoid overpaying relative to player contribution.

Identify Undervalued Players

Target players with strong performance metrics but low market values — the most cost-effective transfer opportunities.

Track Expiring Contracts

Players with contracts ending soon represent lower-cost acquisition targets with reduced transfer fees.

Combine Analytics with Traditional Scouting

Data insights should augment — not replace — traditional scouting for the most robust recruitment decisions.

Optimize Transfer Budgets

Use data-driven allocation models to distribute transfer budgets across positions based on value and need.

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Conclusion

Cleaned Data

Processed 18,979 players across 70+ features into an analysis-ready dataset

EDA Completed

Uncovered patterns in market value, wages, performance, and club composition

Visualizations Built

Six core charts and an interactive Power BI dashboard for stakeholder reporting

Decisions Enabled

Delivered actionable, data-driven transfer and recruitment recommendations

> *"Data transforms football recruitment from intuition into informed decision-making."*

Thank You

Questions are welcome.

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Sumit Shakya

Shubham Satapathy

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